IT-Simplera Institute

Week 1 Task Report

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Network Administration

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Networking Fundamentals & Network Simulation Tools

Local Area Network

Introduction

Networking is the process of connecting devices so they can share data and resources. It is a core part of modern IT systems and forms the backbone of most organizations.

This report presents the work completed in Week 1 of the Network Administration Internship Program at IT-Simplera Institute. The focus of this week was to build a solid understanding of basic networking concepts. This included network devices, IP addressing, and communication models.

To gain practical experience, two widely used simulation tools were studied. These were Cisco Packet Tracer and GNS3. Both tools are used by network professionals to design and test networks in a virtual environment before applying them on real hardware.

A basic LAN topology was built using both tools. Devices were assigned static IP addresses. The ping command was then used to test and confirm connectivity between them. This report explains the steps followed, the concepts understood, and the difficulties faced during the task.

Networking Concepts Learned

Network: A network is a group of devices connected together to share data and resources.

LAN (Local Area Network): A LAN connects devices within a small area, such as an office or a lab. It is used for fast and local communication.

IP Address: An IP address is a unique number assigned to each device on a network. It allows devices to identify and communicate with each other.

Static IP Addressing: In static addressing, an IP address is manually assigned to a device. It does not change unless updated manually.

Network Devices:

- **Router:** Connects different networks and directs data between them.
- **Switch:** Connects multiple devices within the same network and forwards data to the correct device.
- **PC/End Device:** Sends and receives data on the network.

Communication Models: Networking follows standard models like OSI and TCP/IP. These models divide network communication into layers. Each layer has a specific role in sending and receiving data.

Ping Command: Ping is used to test connectivity between two devices. It sends small packets of data and checks if a response is received.

Cisco Packet Tracer Interface Overview

Cisco Packet Tracer is a network simulation tool developed by Cisco. It is used to design, build, and test network topologies in a virtual environment.

Main Parts of the Interface:

Device Panel: Located at the bottom left. It contains categories of network devices like routers, switches, and end devices.

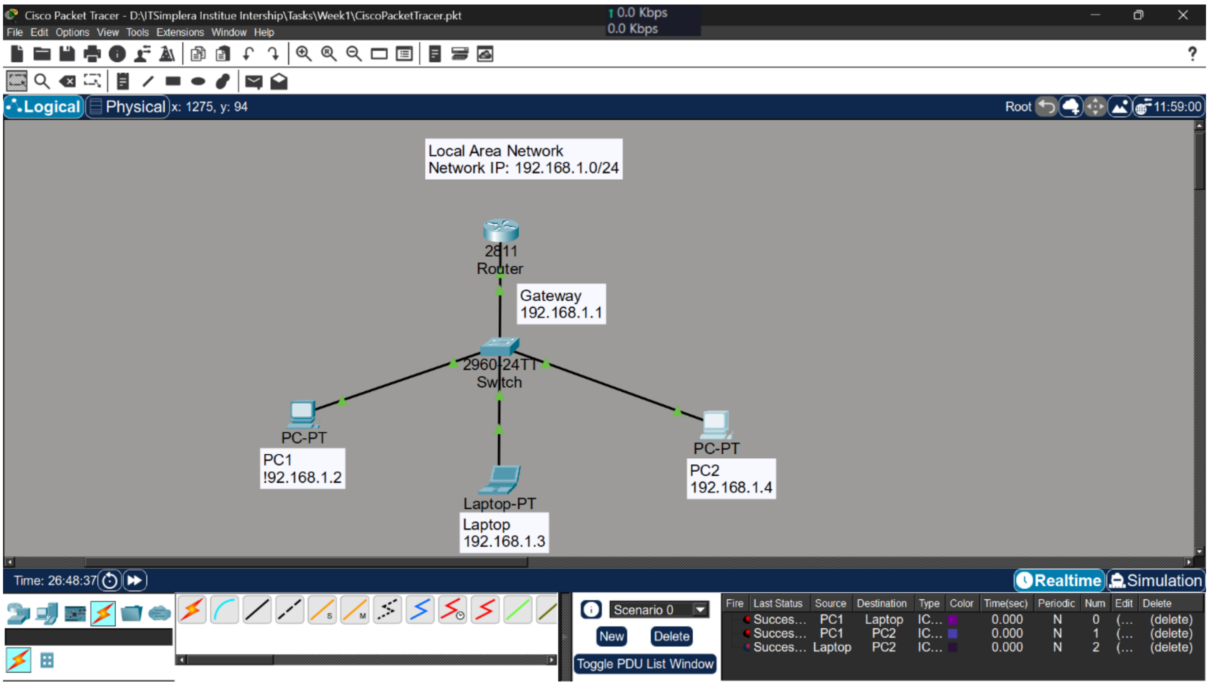
Workspace: The large area in the center. Devices are dragged here to build the topology.

Connections Panel: Contains different types of cables used to connect devices, such as copper straight-through and console cables.

Physical/Logical Toggle: Allows switching between a logical view of the network and a physical view.

Simulation Mode: Used to observe how data packets travel through the network step by step.

Command Line Interface (CLI): Available inside routers and switches. Used to configure devices using commands.



Ping Test Results (Cisco)

PC1: IP:192.168.1.2

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.1.4

Pinging 192.168.1.4 with 32 bytes of data:

Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Laptop: IP: 192.168.1.3

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.1.4

Pinging 192.168.1.4 with 32 bytes of data:

Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

PC2: IP: 192.168.1.4

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time=11ms TTL=128
Reply from 192.168.1.2: bytes=32 time=5ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 11ms, Average = 4ms

C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

GNS3 Interface Overview

GNS3 (Graphical Network Simulator-3) is a network simulation tool used to design and test real network configurations using virtual and emulated devices.

Main Parts of the Interface:

Devices Toolbar: Located on the left side. Contains routers, switches, and end devices that can be added to the project.

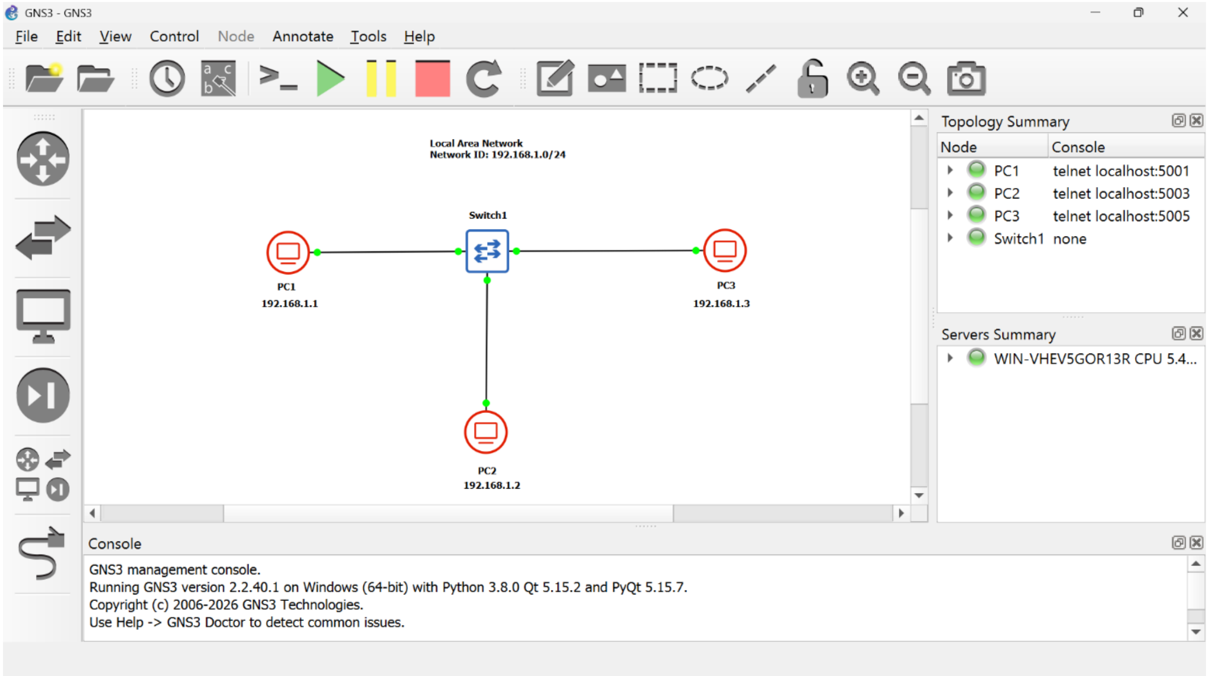
Workspace: The central area where devices are placed and connected to build the topology.

Topology Summary: Shows a list of all devices added to the current project along with their status.

Console Access: Devices are configured through a console connection, usually opened using PuTTY.

Server Summary: Displays information about the local or remote GNS3 server running the simulation.

Start/Stop Controls: Used to start, stop, or restart devices within the project.



Ping Test Results (Gns3)

PC1: IP: 192.168.1.1

```
PC1> ip 192.168.1.1 255.255.255.0
Checking for duplicate address...
PC1 : 192.168.1.1 255.255.255.0

PC1> ping 192.168.1.2
84 bytes from 192.168.1.2 icmp_seq=1 ttl=64 time=1.027 ms
84 bytes from 192.168.1.2 icmp_seq=2 ttl=64 time=1.261 ms
84 bytes from 192.168.1.2 icmp_seq=3 ttl=64 time=1.003 ms
84 bytes from 192.168.1.2 icmp_seq=4 ttl=64 time=1.188 ms
84 bytes from 192.168.1.2 icmp_seq=5 ttl=64 time=1.222 ms

PC1> ping 192.168.1.3
84 bytes from 192.168.1.3 icmp_seq=1 ttl=64 time=1.158 ms
84 bytes from 192.168.1.3 icmp_seq=2 ttl=64 time=1.111 ms
84 bytes from 192.168.1.3 icmp_seq=3 ttl=64 time=1.234 ms
84 bytes from 192.168.1.3 icmp_seq=4 ttl=64 time=1.263 ms
84 bytes from 192.168.1.3 icmp_seq=5 ttl=64 time=1.154 ms

PC1> save
Saving startup configuration to startup.vpc
.  done
```

PC2: IP: 192.168.1.2

```
PC2> ip 192.168.1.2 255.255.255.0
Checking for duplicate address...
PC1 : 192.168.1.2 255.255.255.0

PC2> ping 192.168.1.1
84 bytes from 192.168.1.1 icmp_seq=1 ttl=64 time=1.095 ms
84 bytes from 192.168.1.1 icmp_seq=2 ttl=64 time=1.231 ms
84 bytes from 192.168.1.1 icmp_seq=3 ttl=64 time=1.076 ms
84 bytes from 192.168.1.1 icmp_seq=4 ttl=64 time=1.195 ms
84 bytes from 192.168.1.1 icmp_seq=5 ttl=64 time=1.266 ms

PC2> ping 192.168.1.3
84 bytes from 192.168.1.3 icmp_seq=1 ttl=64 time=1.282 ms
84 bytes from 192.168.1.3 icmp_seq=2 ttl=64 time=1.321 ms
84 bytes from 192.168.1.3 icmp_seq=3 ttl=64 time=1.177 ms
84 bytes from 192.168.1.3 icmp_seq=4 ttl=64 time=1.258 ms
84 bytes from 192.168.1.3 icmp_seq=5 ttl=64 time=1.275 ms

PC2> save
Saving startup configuration to startup.vpc
.  done
```

PC3: IP: 192.168.1.3

```
PC3> ip 192.168.1.3 255.255.255.0
Checking for duplicate address...
PC1 : 192.168.1.3 255.255.255.0

PC3> ping 192.168.1.1
84 bytes from 192.168.1.1 icmp_seq=1 ttl=64 time=1.327 ms
84 bytes from 192.168.1.1 icmp_seq=2 ttl=64 time=1.191 ms
84 bytes from 192.168.1.1 icmp_seq=3 ttl=64 time=1.174 ms
84 bytes from 192.168.1.1 icmp_seq=4 ttl=64 time=0.986 ms
84 bytes from 192.168.1.1 icmp_seq=5 ttl=64 time=1.262 ms

PC3> ping 192.168.1.2
84 bytes from 192.168.1.2 icmp_seq=1 ttl=64 time=1.287 ms
84 bytes from 192.168.1.2 icmp_seq=2 ttl=64 time=2.246 ms
84 bytes from 192.168.1.2 icmp_seq=3 ttl=64 time=1.208 ms
84 bytes from 192.168.1.2 icmp_seq=4 ttl=64 time=1.144 ms
84 bytes from 192.168.1.2 icmp_seq=5 ttl=64 time=1.423 ms

PC3> sav
Saving startup configuration to startup.vpc
.  done
```

Challenges Faced

I was already familiar with Cisco Packet Tracer from previous coursework and lab work. Because of this, I did not face any major difficulty while building the topology or configuring the devices in Packet Tracer.

GNS3 was completely new to me. I had never used it before this task. I started by exploring and understanding its interface, since it is quite different from Packet Tracer. Once I became comfortable with the layout, I learned how to add and connect switches and client PCs using links.

I then learned how to access devices through the CLI and configure IP addressing on the end devices. After the configuration, I performed ping tests to verify connectivity between the devices. Since everything was new, it took some time to get used to the workflow, but I was able to complete the task by exploring the tool step by step.

Conclusion

This week helped me build a solid foundation in basic networking concepts, including IP addressing, network devices, and communication models. I gained hands-on experience with both Cisco Packet Tracer and GNS3 by building a simple LAN topology in each tool. I was already comfortable with Packet Tracer, but learning GNS3 for the first time improved my understanding of network simulation tools and CLI-based device configuration. Overall, this task strengthened my practical networking skills and prepared me for more advanced tasks ahead.
